Logic, Language, and Philosophy

Soundness and completeness

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Overview

Background

Proposition Logic: Soundness

Proposition Logic: Completeness

Homework

Background

Validity and its explications

Semantic validity and syntactic derivability both aim to explain *intuitive validity*.

Observations:

- Syntactic derivability implies intuitive validity:
  we can see that all the steps in a derivation are intuitively correct
- Intuitively validity implies semantic validity:
  if an argument is semantically not valid, then it is not intuitively valid: a counter-example convinces us of that

Validity and its explications (cont.)

Remaining questions:

- Is syntactic derivability not too restricted, i.e., is everything that is intuitively valid also syntactically derivable?
- Is semantic validity not too broad, i.e., is everything that is semantically valid also intuitively valid?

Together, soundness and completeness answer these questions positively: the three notions coincide.

Meta-theorems

The soundness and completeness theorems are so-called ‘meta-theorems’, i.e., not theorems *in* (or *of*) a logical system, but *about* such a system.

Meta-theorems are based on reasoning about properties of a logical system, such as its semantics and its proof theory.

Meta-theorems: semantics

In the semantics of proposition logic and of predicate logic we have already encountered examples of meta-theorems.

Fact

If $\phi \to \psi$ is a contradiction, then ϕ is a tautology and ψ is a contradiction

Fact

$\{\dagger\}$ is a functionally complete set of connectives

Fact

If $\models \phi$ and $\models \phi \to \psi$, then $\models \psi$

Fact

$\phi \leftrightarrow \psi \models \chi \leftrightarrow [\phi/\psi]\chi$ (Extensionality)

Meta-theorems: deduction

We can also formulate and prove meta-theorems about proofs:

Fact (1)

If $\phi \vdash \psi$ and $\psi \vdash \chi$, then $\phi \vdash \chi$

Fact (2)

If $\neg\phi \vdash \phi$, then $\vdash \phi$

Fact (3)

If $\Gamma \vdash \psi$, then $\phi_1 \dots \phi_n \vdash \psi$ for a finite set $\{\phi_1 \dots \phi_n\} \subseteq \Gamma$

Fact (4)

Suppose $\phi_1 \dots \phi_n \vdash \psi$. Then there is a derivation of ψ that starts with the assumptions $\phi_1 \dots \phi_n$ in which after step n no assumptions occur that are not withdrawn.

Fact 1

If $\phi \vdash \psi$ (1) and $\psi \vdash \chi$ (2) , then $\phi \vdash \chi$ (3)

$$(1) \begin{cases} \phi & \mathtt{A} \\ \Pi_1 & \\ \psi & \end{cases}$$

and

$$(2) \begin{cases} \psi & \mathtt{A} \\ \Pi_2 & \\ \chi & \end{cases}$$

assemble into:

$$(3) \begin{cases} \phi & \mathtt{A} \\ \Pi_1 & \\ \psi & \\ \Pi_2 & \\ \chi & \end{cases}$$

Fact 2

If $\neg\phi \vdash \phi$, then $\vdash \phi$

$$
\begin{array}{|ll}
\neg\phi & \mathtt{A} \\
\Pi_1 & \\
\phi & \\
\bot & E_\neg \\
\hline
\neg\neg\phi & I_\neg \\
\phi & \neg\neg - \text{rule}
\end{array}
$$

Of facts, lemmas, and theorems

Simple results, such as the above, are usually called *facts*.

More complex results that are used to prove theorems are often referred to as *lemmas*.

The term *theorem* is reserved for main results.

An example is the Deduction Theorem, which states a connection between derivability and implication.

Deduction theorem

Theorem (Deduction theorem)

If $\Gamma \cup \{\phi\} \vdash \psi$, *then* $\Gamma \vdash \phi \to \psi$

Proof: Suppose $\Gamma \cup \{\phi\} \vdash \psi$. Then there is a proof of the following form:

$$\begin{array}{ll} \Pi_0 & \\ \phi & \text{ass.} \\ \Pi_1 & \\ \psi & \end{array}$$

in which Π_0 contains only assumptions and Π_1 does not contain any assumptions that are not withdrawn (by Fact 4). It is easy to see that it follows from this that $\Gamma \vdash \phi \to \psi$: simply continue the proof with $I_\to$:

$$\begin{array}{|ll} \Pi_0 & \\ \phi & \text{ass.} \\ \Pi_1 & \\ \psi & \\ \hline \phi \to \psi & I_\to \end{array}$$

■

Cut-rule

Theorem (Cut-rule)

If $\Gamma \vdash \phi$ *and* $\Gamma \cup \{\phi\} \vdash \psi$*, then* $\Gamma \vdash \psi$

- Let Π_0 be the active assumptions from Γ and let Π_1 consist of the remainder of the derivation of ϕ from Γ
- Let Π_2 be the active assumptions from Γ and let Π_3 consist of the remainder of the derivation of ψ from $\Gamma \cup \phi$

$$(1) \begin{cases} \Pi_0 \\ \Pi_1 \\ \phi \end{cases}$$

$$(2) \begin{cases} \Pi_2 \\ \Pi_3 \\ \psi \end{cases}$$

Proposition Logic: Soundness

Soundness

The *soundness theorem* for proposition logic states that if ϕ is derivable in the system of natural deduction from a set of premisses Γ, then ϕ is also semantically entailed by Γ.

To prove this we first ascertain that every rule of the natural deduction system as such is sound. The proof of the soundness theorem then follows easily.

Example: $E_{\rightarrow}$

$E_{\rightarrow}$-rule:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m_1. & \phi \rightarrow \psi & \\
\vdots & & \\
m_2. & \phi & \\
\vdots & & \\
k. & \psi & E_{\rightarrow}, m_1, m_2
\end{array}
$$

Fact (5)

If $\Gamma \models \phi$ *and* $\Gamma \models \phi \rightarrow \psi$, *then* $\Gamma \models \psi$

Proof: Suppose that $\Gamma \models \phi$ and $\Gamma \models \phi \rightarrow \psi$. Take V such that $V(\chi) = 1$, for all $\chi \in \Gamma$. From $\Gamma \models \phi$ it follows that $V(\phi) = 1$. $\Gamma \models \phi \rightarrow \psi$ implies that $V(\phi \rightarrow \psi) = 1$. From this it follows directly that $V(\psi) = 1$.

Second example: $E_\neg$

$E_\neg$-rule:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & \neg\phi & \\ \vdots & & \\ k. & \phi & \\ \vdots & & \\ n. & \bot & E_\neg, m, k \end{array}$$

Fact

If $\Gamma \models \phi$ *and* $\Gamma \models \neg\phi$*, then* $\Gamma \models \bot$

Proof goes by contraposition: if $\Gamma \not\models \bot$ then it is not the case that $\Gamma \models \phi$ and $\Gamma \models \neg\phi$

- Suppose $\Gamma \models \phi$ and $\Gamma \models \neg\phi$
- Then $\Gamma \models \bot$, so for any V: $V(\gamma) = 1$ for all $\gamma \in \Gamma$, we would have $V(\bot) = 1$
- Contradiction: per definition there is no such V
- So: there is no V: $V(\gamma) = 1$ for all $\gamma \in \Gamma$ and $V(\phi) = V(\neg\phi) = 1$
- Hence either $\Gamma \not\models \phi$ or $\Gamma \not\models \neg\phi$ (or both)

The Soundness Theorem

Theorem (Soundness Theorem)

If $\Gamma \vdash \phi$, *then* $\Gamma \models \phi$

Outline of the proof:

We prove the following:

$$\text{For every } \phi \text{ and } \Gamma: \text{ if } \Pi \text{ is a deduction of } \phi \text{ from } \Gamma, \text{ then } \Gamma \models \phi$$

The proof is by induction on the number of steps in the deduction Π.

Base step

Base step:

Π consists of one step. Then Π is of the form :

$$1. \quad \phi \quad \text{ass.}$$

and hence ϕ is an element of Γ. Obviously, every V that makes Γ true also makes ϕ true.

Induction step

Induction step:

Π consists of more than one step. The induction hypothesis states that soundness has already been proven for all derivations that are a proper part of Π. What needs to be shown is that the last step in Π is sound. We do that by going through all the cases.

Case: $E_\wedge$

$E_\wedge$. Suppose ϕ is derived using $E_\wedge$. Then Π is of the following form:

$$\begin{array}{ll} \Pi_0 & \\ \phi \wedge \psi & \\ \Pi_1 & \\ \phi & E_\wedge \end{array}$$

in which Π_0 consists of the active assumptions from Γ and Π_1 has no active assumptions. The induction hypothesis tells us that $\Gamma \models \phi \wedge \psi$. We have as a fact that if $\Gamma \models \phi \wedge \psi$, then $\Gamma \models \phi$ and it then follows that $\Gamma \models \phi$. ■

Second case: $I_\to$

$I_\to$. Suppose ψ is derived using $I_\to$. Then Π is of the following form

$$\begin{array}{|ll} \Pi_0 & \\ \phi & \mathtt{A} \\ \Pi_1 & \\ \psi & \\ \hline \phi \to \psi & I_\to \end{array}$$

The induction hypothesis tells us:

$$\begin{array}{ll} \Pi_0 & \\ \phi & \mathtt{A} \\ \Pi_1 & \\ \psi & \end{array}$$

is sound.

So: $\Gamma \cup \{\phi\} \models \psi$, from that it follows that $\Gamma \models \phi \to \psi$

Using soundness

Theorem (Soundness Theorem, contraposition)

If $\Gamma \not\models \phi$, *then* $\Gamma \not\vdash \phi$

This can be used to prove non-derivability via specification of a semantic counter-example.

Example: in order to show that $p \to q \not\vdash q \to p$ it suffices to show that $p \to q \not\models q \to p$, which we do by specifying a valuation V for p and q such that $V(p \to q) = 1$ and $V(q \to p) = 0$.

Proposition Logic: Completeness

Completeness theorem: direct formulation

Theorem (Completeness Theorem (direct))

If $\Gamma \models \phi$, *then* $\Gamma \vdash \phi$

Observations:

- direct proof is do-able for proposition logic
- more difficult for predicate logic:
  - entailment (validity) quantifies over a potentially infinite collection of potentially infinite models

Completeness theorem: indirect formulation

Prove completeness via contraposition (the 'Henkin'-method):

Theorem (Completeness Theorem (indirect))

If $\Gamma \nvdash \phi$, *then* $\Gamma \not\models \phi$

Consistency and satisfiability

Two key notions:

- *Consistency*: Γ is consistent iff $\Gamma \nvdash \bot$
- *Satisfiability*: Γ is satisfiable if there is a valuation V such that $V(\phi) = 1$ for every $\phi \in \Gamma$

Auxiliary fact:

Fact

If $\Gamma \nvdash \phi$, then $\Gamma \cup \{\neg\phi\}$ is consistent

Note:

- Auxiliary fact links derivability with consistency
- If we can then link consistency with satisfiability, we can prove completeness

Fact 11

First Fact 1:
If $\Gamma \nvdash \phi$, then $\Gamma \cup \{\neg\phi\}$ is consistent

Proof:

- Contraposition: If $\Gamma \cup \{\neg\phi\}$ is *in*consistent, then $\Gamma \vdash \phi$
- Suppose $\Gamma \cup \{\neg\phi\}$ is *in*consistent
- Then: $\Gamma \cup \{\neg\phi\} \vdash \bot$ (by definition of consistency)
- That means that we have:

$$
\begin{array}{|ll}
\Gamma & \\
\hline
\neg\phi & \mathbf{A} \\
\Pi_0 & \\
\bot & \\
\hline
\neg\phi \to \bot & I_{\to} \\
\hline
\neg\phi & \mathbf{A} \\
\bot & \\
\hline
\neg\neg\phi & I_{\neg} \\
\phi &
\end{array}
$$

- which shows that $\Gamma \vdash \phi$

The consistency theorem

So:

> *Fact 1 links derivability with consistency*

Then:

> *If consistency implies satisfiability, then we have completeness*

So, proof of the indirect formulation of the completeness theorem boils down to proof of:

Theorem (Consistency Theorem (Henkin))

If Γ is consistent, then there is a V: $V(\phi) = 1$ for every $\phi \in \Gamma$

Groundwork

How to do that?

- Basic idea: show that you can construct a valuation that satisfies Γ if and only if Γ is consistent
- Note: a valuation assigns a truth value to *all* formulas
- So we need to construct a consistent and 'complete' set of formulas from a consistent set

Such a 'complete' set of formulas Γ is called 'maximally consistent'

Required:

> *Construction that extends any consistent* Γ *into a maximally consistent* Γ'

Consistent and maximally consistent sets

Definition (Consistency)

A set of formulas Γ is *consistent* iff $\Gamma \nvdash \bot$, otherwise Γ is *inconsistent*

Definition (Maximal consistency)

A set of formulas Γ is *maximally consistent* iff

1. Γ is consistent
2. every Γ' such that $\Gamma' \supset \Gamma$ is inconsistent

So, Γ is maximally consistent iff any addition of a formula ϕ makes it inconsistent

Maximally consistent sets and valuations

Key insight:

Fact

A maximally consistent set defines a unique valuation

Syntactically closed sets

Two properties that are connected with (maximal) consistency are the following:

- Γ is *syntactically closed* iff Γ is consistent and for every ϕ it holds that $\Gamma \vdash \phi$ or $\Gamma \vdash \neg\phi$
- Γ is *deductively closed* iff if $\Gamma \vdash \phi$ then $\phi \in \Gamma$

Some facts about consistency

Fact

1. Γ *is consistent iff there is at least one formula* ϕ *such that* $\Gamma \nvdash \phi$
2. $\Gamma \cup \{\phi\}$ *is inconsistent iff* $\Gamma \vdash \neg\phi$
3. $\Gamma \cup \{\neg\phi\}$ *is consistent iff* $\Gamma \nvdash \phi$
4. *If* Γ *is consistent, at least one of the two sets* $\Gamma \cup \{\phi\}$, $\Gamma \cup \{\neg\phi\}$ *is also consistent*

Ad 1

1. Γ is consistent iff there is at least one formula ϕ such that $\Gamma \nvdash \phi$

Proof:

- $\Rightarrow$, i.e., suppose Γ consistent:
  - suppose $\Gamma \vdash \phi$ for all ϕ
  - then $\Gamma \vdash \phi$ and $\Gamma \vdash \neg\phi$
  - but then $\Gamma \vdash \bot$, and hence Γ inconsistent
  - contradiction, so it is not the case that $\Gamma \vdash \phi$ for all ϕ
- $\Leftarrow$, i.e., suppose $\Gamma \nvdash \phi$:
  - suppose Γ not consistent
  - then $\Gamma \vdash \bot$ (definition consistency)
  - with EFSQ we can then conclude that $\Gamma \vdash \phi$
  - contradiction, so it is not the case that Γ is not consistent

Ad 3

3. $\Gamma \cup \{\neg\phi\}$ is consistent iff $\Gamma \nvdash \phi$

Proof:

- $\Rightarrow$, i.e., suppose $\Gamma \cup \{\neg\phi\}$ is consistent
  - suppose $\Gamma \vdash \phi$
  - then we have:

$$\begin{array}{ll} \Pi_0 & \\ \neg\phi & \\ \phi & \\ \bot & \end{array}$$

  - contradiction
- $\Leftarrow$, i.e., suppose $\Gamma \vdash \phi$
  - then we have:

$$\begin{array}{lr} \Pi_0 & \\ \phi & \\ \neg\phi & \mathtt{A} \\ \bot & \end{array}$$

  - contradiction

Some facts about maximal consistency

Fact (Maximal consistent sets)

Let Γ be a maximally consistent set, and ϕ and ψ arbitrary formulas. Then:

1. $\phi \in \Gamma$ *iff* $\Gamma \vdash \phi$
2. $\neg\phi \in \Gamma$ *iff* $\phi \notin \Gamma$
3. $\phi \wedge \psi \in \Gamma$ *iff* $\phi \in \Gamma$ *and* $\psi \in \Gamma$
4. $\phi \vee \psi \in \Gamma$ *iff* $\phi \in \Gamma$ *or* $\psi \in \Gamma$
5. $\phi \rightarrow \psi \in \Gamma$ *iff* $\phi \notin \Gamma$ *or* $\psi \in \Gamma$

Note that 1 states that maximally consistent sets are deductively closed, and that from 2 in combination with 1 we can conclude that such sets are also syntactically closed.

Ad 1

1. $\phi \in \Gamma$ iff $\Gamma \vdash \phi$

Proof:

- $\Rightarrow$: trivial
- $\Leftarrow$:
  - suppose $\Gamma \vdash \phi$ and $\phi \notin \Gamma$
  - then adding ϕ to Γ would make Γ inconsistent, because Γ is *maximally* consistent
  - i.e., we then would have $\Gamma \cup \{\phi\} \vdash \bot$, which means:

$$\begin{array}{ll} \Pi_0 & \\ \phi & \mathtt{A} \\ \bot & \end{array}$$

  - but we also have $\Gamma \vdash \phi$, which means:

$$\begin{array}{l} \Pi_0 \\ \phi \end{array}$$

  - so we have:

$$\begin{array}{l} \Pi_0 \\ \bot \end{array}$$

  - which means that Γ is inconsistent
  - contradiction

Ad 3

3. $\phi \wedge \psi \in \Gamma$ iff $\phi \in \Gamma$ and $\psi \in \Gamma$

Proof:

- $\Rightarrow$, i.e., $\phi \wedge \psi \in \Gamma, \phi \notin \Gamma$
  - if $\phi \notin \Gamma$, then $\Gamma \cup \{\phi\}$ inconsistent
  - so $\Gamma \cup \{\phi\} \vdash \bot$
  - which means $\Gamma \vdash \phi \to \bot$, which means $\Gamma \vdash \neg\phi$
  - now $\phi \wedge \psi \in \Gamma$ means $\Gamma \vdash \phi \wedge \psi$, and thus $\Gamma \vdash \phi$
  - so, from Γ both ϕ and $\neg\phi$ would be derivable
  - contradiction
- $\Leftarrow$, i.e., $\phi, \psi \in \Gamma$:
  - then $\Gamma \vdash \phi$ and $\Gamma \vdash \psi$, hence $\Gamma \vdash \phi \wedge \psi$
  - suppose $\phi \wedge \psi \notin \Gamma$
  - then $\Gamma \cup \{\phi \wedge \psi\} \vdash \bot$
  - but then $\Gamma \vdash \bot$
  - contradiction

Lindenbaum's Theorem

The key step, getting from consistent to maximally consistent sets. This can be done because of:

Lemma (Lindenbaum)

Every consistent set of formulas is contained in a maximally consistent set.

Construction

Let $\phi_0 \ldots \phi_i \ldots$ be an ordering of all the formulas.

Let Γ be a consistent set.

Then construct Γ' as follows:

$\Gamma' = \bigcup_n \Gamma_n$, where:

- $\Gamma_0 = \Gamma$
- $\Gamma_{n+1} = \begin{cases} \Gamma_n \cup \{\phi_n\} \text{ if the result is consistent} \\ \Gamma_n \text{ otherwise} \end{cases}$

Claim: Γ' is maximally consistent

Consistency Theorem: outline of proof

Theorem (Consistency theorem (Henkin))

If Γ is consistent, then there exists a valuation V such that $V(\phi) = 1$, for all $\phi \in \Gamma$

Proof: We know that every consistent Γ is contained in a maximally consistent Γ' (Lindenbaum). Choose a valuation V such that $V(p) = 1$ iff $p \in \Gamma'$. With induction on the construction of ϕ we prove that $V(\phi) = 1$ iff $\phi \in \Gamma'$.

Base step

Base step: ϕ is an atomic proposition p:
If $p \neq \bot$, then the result immediately follows;
If $p = \bot$, then $V(p) = 0$ and $p \notin \Gamma'$, by the consistency of Γ'. ■

Induction step: examples

Induction step (examples)

- $\phi = \neg\psi$:
  $V(\neg\psi) = 1$ iff $V(\psi) = 0$ (truth definition)
  So $\psi \notin \Gamma'$ (induction hypothesis)
  $\neg\phi \in \Gamma$ iff $\phi \notin \Gamma$ ((fact 2)
  Thus $\neg\psi \in \Gamma'$
- $\phi = \psi \to \chi$:
  $V(\psi \to \chi) = 1$ iff $V(\psi) = 0$ or $V(\chi) = 1$ (truth definition)
  So $\psi \notin \Gamma'$ or $\chi \in \Gamma'$ (induction hypothesis)
  $\phi \to \psi \in \Gamma$ iff $\phi \notin \Gamma$ or $\psi \in \Gamma$ (fact 5)
  Thus $\psi \to \chi \in \Gamma'$
- $\phi = \psi \wedge \chi$:
  $V(\psi \wedge \chi) = 1 \Leftrightarrow V(\psi) = V(\chi) = 1$ (truth definition)
  So $\psi, \chi \in \Gamma'$ (induction hypothesis)
  $\phi \wedge \psi \in \Gamma$ iff $\phi \in \Gamma$ and $\psi \in \Gamma$ (fact 3)
  Thus $\psi \wedge \chi \in \Gamma'$

Homework

Homework

- Read chapter 20 and Engel's paper 'Logic, Reasoning and the Logical Constants'
- Read chapter 21 and make exercises 1(3), 2(4,8), 3